

1 Metric spaces

Problem 1. Let X be a set with two members: a and b . Define $d(a, a) = d(b, b) = 0$, $d(a, b) = d(b, a) = r$, where r is a positive real number. Prove that X is a metric space relative to the function d .

Problem 2. Let X be a set with three elements: a, b , and c . Define d so that $d(x, x) = 0$ for all x , and $d(x, y) = d(y, x)$ is a positive real number for $x \neq y$. Say $d(a, b) = r$, $d(a, c) = s$, $d(b, c) = t$ with $r \leq s \leq t$. Prove that d makes X a metric space if and only if $t \leq r + s$.

Problem 3. On an arbitrary set X suppose that $d(x, x) = 0$, and for $x \neq y$, $d(x, y) = d(y, x)$ is a number between 1 and 2 (the end points 1 and 2 are permitted). Prove that X is a metric space.

Problem 4. Let X be a set with a real function d satisfying $d(a, a) = 0$, $d(a, b) \neq 0$ for $a \neq b$, and $d(a, b) + d(b, c) \geq d(c, a)$. Prove that d makes X a metric space. (Note: $d(a, b) \geq 0$ and $d(a, b) = d(b, a)$ are not being assumed; they must be proved.)

Problem 5. In any metric space, prove that $d(a_1, a_n) \leq d(a_1, a_2) + d(a_2, a_3) + \cdots + d(a_{n-1}, a_n)$.

Problem 6. A metric space X is called ultra-metric if $d(a, c) \leq \max[d(a, b), d(b, c)]$ for all $a, b, c \in X$. Show that discrete metric is ultra-metric. Is the real line with euclidean metric a ultra-metric?

Problem 7. Let $f(x)$ be a real-valued function of a real variable, defined for all $x \geq 0$. Then f is said to be concave if for any nonnegative real numbers a, b with $a + b = 1$ we have

$$f(ax + by) \geq af(x) + bf(y)$$

Suppose that f is concave, that $f(0) = 0$, that $f(x) > 0$ for $x > 0$, and that f is monotone in the sense $x \leq y$ implies $f(x) \leq f(y)$. Let (X, d) be a metric space, then prove that $(X, f(d))$ is also a metric space.

Problem 8. If f (a real function of a real variable) has a second derivative satisfying $f'' \leq 0$, prove that f is concave.

Problem 9. By using the above problems or otherwise, prove the following: If d is a metric on any set X , then so is $f(d)$; for $f(x) = \sqrt{x}$, $f(x) = \frac{x}{1+x}$, and $f(x) = \min(x, 1)$.

Problem 10. If d is a metric on X , is d^2 necessarily a metric?

Problem 11. Let d_1 and d_2 be metrics on a space X . Which of the following are metrics on X ; $d_1 + d_2$, $\max(d_1, d_2)$, $\min(d_1, d_2)$?

Problem 12. Let (X, d) be a metric space in which the distance function d assumes only the values 0, 1, 3. Define $x \sim y$ to mean $d(x, y) \leq 1$. Prove that $\sim$ is an equivalence relation.

Problem 13. Let A be any set of positive real numbers. Prove that there exists a metric space whose nonzero distances constitute exactly the set A .

Problem 14. ("Antimetric spaces.") Let d on a set X satisfy the axioms for a metric space except that the triangle inequality is reversed:

$$d(a, c) \geq d(a, b) + d(b, c)$$

Prove that X has at most one point.

Problem 15. Let (X_1, d_1) and (X_2, d_2) be metric spaces, then a function $f : X_1 \rightarrow X_2$ is said to be an isometry from X_1 to X_2 if

$$d_2(f(a), f(b)) = d_1(a, b), \quad \forall a, b \in X_1.$$

Give an example of a metric space (X, d) which admits an isometry with a proper subspace (Q, d) of itself (i.e. $Q \subsetneq X$). (Hint: Use discrete metric.)

Problem 16. (a) Suppose that a metric space A is a union $A = B \cup C$ of two subsets of finite diameter. Prove A has finite diameter.

(b) Generalize part (a) to the union of a finite number of subsets.

(c) Suppose that, in part (a), $B \cap C$ is nonempty. Prove that $\text{diam}(A) \leq \text{diam}(B) + \text{diam}(C)$.

2 Open Sets

Problem 17. Prove that any set in a metric space is an arbitrary intersection of open sets.

Problem 18. If a metric space (X, d) has the property that any arbitrary intersection of open sets is open, prove that (X, d) is discrete.

Problem 19. Let M be a finite open subset of a metric space (X, d) . Then prove that every point in M is an isolated point of (X, d) .

Problem 20. Given a point $x_0 \in X$ in a metric space (X, d) , and positive numbers r, s with $r < s$, prove that the following set is open

$$\{x \in X : r < d(x, x_0) < s\}.$$

Problem 21. (DIFFICULT) Let (X, d) be an infinite metric space (i.e. $\text{card}(X) \not\leq \aleph_0$). Prove that there exist a $U \subseteq X$ contains an open set U such that

$$\text{card}(U) \not\leq \aleph_0 \text{ and } \text{card}(U^c) \not\leq \aleph_0.$$

Problem 22. (DIFFICULT) Let (X, d) be an ultra metric space. If $A, B \subseteq X$ are open sets such that $A \cap B \neq \emptyset$ then show that

$$\text{either } A \subseteq B \text{ or } B \subseteq A.$$

Problem 23. Let (X, d) be metric space and let τ the set of all open subsets of X . If τ is countable then show that X is also countable.

3 closed set and sequences

Problem 24. If in a metric space a sequence converges to both x and y , prove that $x = y$.

Problem 25. Prove that in a metric space (X, d) , $x_n \xrightarrow{d} x$ if and only if $d(x_n, x) \rightarrow 0$.

Problem 26. Let (X, d) be a metric space, and let $A, B \subseteq X$. Then show that $\overline{A \cup B} = \overline{A} \cup \overline{B}$. Does $\overline{A \cap B} = \overline{A} \cap \overline{B}$ always hold?

Problem 27. Given distinct points x and y in a metric space, prove that there exist open sets U and V such that $x \in U, y \in V$, and $\bar{U} \cap \bar{V} = \emptyset$.

Problem 28. Let (X, d) be a metric space and let $A \subseteq X$. Then show that

$$\text{diam}(A) = \text{diam}(\bar{A}).$$

Problem 29. Let $M \subseteq X$ be a subset of a metric space (X, d) .

(a) If U is an open set in the subspace (X, d) , then prove that $U \cap M$ is open in the metric space (M, d) .

(b) If F is a closed set in (X, d) , prove that $F \cap A$ is closed in (M, d) .

(c) Prove that if O (resp. C) is any open (resp. closed) set in (M, d) then there exists an open set U (resp. closed E) set in (X, d) such that

$$O = U \cap M \quad (\text{resp.}) \quad C = E \cap M.$$

Problem 30. Give an example where A and B are both closed in some metric space (X, d) , such that $A \cap B = \emptyset$ is empty, and $\inf_{x,y} d(x, y) = 0$. (Suggestion: a hyperbola and its asymptotes.)

Problem 31. Let (X, d) be a metric space in which the closure of every open set is open. Prove that (X, d) is the discrete metric space.

Problem 32. Difficult Show that in a metric space every open set is the union of a countable number of closed sets, and every closed set is the intersection of a countable number of open sets.

Problem 33. Prove that a metric space is discrete if and only if every convergent sequence is eventually constant.

4 Boundary, interior and Closure

Problem 34. Let (X, d) be a metric space, and let $A \subseteq X$. Then verify the following identities

- (i) $A^\circ \subseteq A \subseteq \overline{A}$. Find example such that the containment are strict.
- (ii) $\partial A = \overline{A} \setminus A^\circ$
- (iii) $\partial A = \partial(A^C)$
- (iv) $\overline{A} = (A^\circ) \cup \partial A$
- (v) $\partial A = \emptyset \iff A$ is both open and closed
- (vi) $\partial(\partial A) \subseteq \partial A$.
- (vii) Show that $\partial(\overline{A}) \subseteq \partial(A)$.

Problem 35. Find example of a subset A of a metric space such that

$$\overline{A^\circ} \subsetneq \overline{A},$$

Problem 36. Find example such that $A^\circ \subsetneq \overline{A}^\circ$

Problem 37. Let $X = \{0, 1\}^\mathbb{N}$, that is $X = \{x = \{x_n\}_{n \in \mathbb{N}} : x_n = 0 \text{ or } x_n = 1\}$. Define metrics on X

$$d_m(x, y) = \begin{cases} 0 & \text{if } x = y \\ 2^{-\min(n) : x_n \neq y_n} & \text{for } x \neq y; \end{cases}$$

and

$$d_H(x, y) = \sum_{k=1}^{\infty} \frac{|x_k - y_k|}{2^k}.$$

Show that $U \subset X$ is open in (X, d_m) if and only if U is open in (X, d_H) .

Problem 38. Show that d_m and d_H defined above are not equivalent metric.

Problem 39.

5 Compact Sets, completeness, and continuous function

In this section you can assume that $(\mathbb{R}, |\cdot|)$ is complete. And $\mathbb{Q}$ is not complete in $(\mathbb{R}, |\cdot|)$. And $\overline{\mathbb{Q}} = \mathbb{R}$.

Problem 40. Let f be a function from a metric space to a second metric space. Prove that f is continuous if and only if it sends convergent sequences into convergent sequences.

Problem 41. Let $u \in X$ be a fixed point of a metric space (X, d) . The function $f(x) = d(u, x)$ maps X into the real numbers. Prove that f is continuous.

Problem 42. Let $\emptyset \neq A \subseteq X$ be a fixed subset of a metric space (X, d) . The function $f(x) = d(A, x) = \inf_{a \in A} d(a, x)$ maps X into the real numbers. Prove that f is continuous.

Problem 43. Let A be a closed subset and y a point in a metric space (X, d) . If $y \in X \setminus A$ then prove that there exists a continuous real function on X which vanishes on A but not at y .

Problem 44. Let (X_1, d_1) and (X_2, d_2) be metric spaces, f a function from X_1 to X_2 .

- (a) If $X_1 = \cup_{i \in I} U_i$ where each U_i is a open set. If f is continuous on each U_i , then prove that f is continuous on X .
- (b) If X is a finite union of closed sets $F_1, \dots, F_n$ on each of which f is continuous, prove that f is continuous on X .
- (c) Can part (b) be extended to an infinite number of closed sets?

Problem 45. Prove that a metric space X is discrete if and only if every function on X (to an arbitrary second metric space) is continuous.

Problem 46. If $f : X \rightarrow Y$ is continuous at x_0 , and $g : Y \rightarrow Z$ is continuous at $f(x_0)$, prove that $g \circ f : X \rightarrow Z$ is continuous at x_0 .

Problem 47. Two metric spaces are said to be homeomorphic if there exists between them a one-to-one correspondence which is continuous in both directions. Prove that any metric space is homeomorphic to one of finite diameter. Hint: Use:

$$x_n \xrightarrow{d} x \Leftrightarrow x_n \xrightarrow{\frac{d}{1+d}} x.$$

Problem 48. (a) A mapping from one metric space to a second is called open if it sends open sets into open sets. Give an example of a continuous function which is not open and an open function which is not continuous.

(b) Similarly a mapping is called closed if it sends closed sets into closed sets. Give examples, as in part (a), to distinguish closed mappings from continuous ones.

(c) Let (X_1, d_1) and (X_2, d_2) be metric spaces, and let f be a function from X_1 to X_2 which is one-to-one and onto. Prove that the following three statements are equivalent:

- (i) f is open.
- (ii) f is closed.
- (iii) f^{-1} is continuous.

Problem 49. Let $\ell^\infty(\mathbb{N}) := \{(x_n)_{n \in \mathbb{N}} : \sup_{n \in \mathbb{N}} |x_n| < \infty\}$. Define a metric on $\ell^\infty(\mathbb{N})$ as

$$d((x_n), (y_n)) = \sup_{n \in \mathbb{N}} |x_n - y_n|.$$

Show that (ℓ^∞, d) is complete.

Problem 50. Let $A \subseteq X$ be dense in (X, d) , that is $\overline{A} = X$. Suppose that every Cauchy sequence in A converges to a point in X . Then prove that (X, d) is complete.

Problem 51. Let (X_1, d_1) and (X_2, d_2) be metric spaces. Assume that X_1 is complete and that there exists $f : X_1 \rightarrow X_2$ which is one-to-one, onto, and continuous. Suppose that f^{-1} is uniformly continuous. Prove that X_2 is complete.

Problem 52. If every countable closed subset of a metric space M is complete, prove that M is complete.

Problem 53. If every closed ball of a metric space (X, d) is complete, prove that (X, d) is complete. Recall, every closed ball is of this form

$$\overline{B_r(x)} = \{y \in X : d(x, y) \leq r\}.$$

In fact it is a good exercise to show that both the sets above are same.

Problem 54. Let A and B be subsets of a metric space. If A and B are complete, prove that $A \cup B$ and $A \cap B$ are complete.

Problem 55. If a is a fixed point in a metric space (X, d) , prove that $d(a, x)$ is a uniformly continuous function from X to the real numbers.

Problem 56. If $\emptyset \neq A$ is a fixed subset of a metric space (X, d) , then prove that $d(A, x)$ is a uniformly continuous function from X to the real numbers.

Problem 57. Prove that the following three statements are equivalent for a metric space (X, d)

1. Every Cauchy sequence in (X, d) is eventually constant.
2. (X, d) is complete and discrete.
3. Every subset of X is complete.

Problem 58. In the metric space of real numbers, give an example of a descending sequence of nonempty closed sets with empty intersection.

Problem 59. The following functions are continuous from the real numbers to the real numbers. Which are uniformly continuous?

(a) x^2 .

(b) $|x|$.

(c) $\frac{1}{1+x^2}$.

Problem 60. Let $(\mathbb{R}, |\cdot|)$ be the metric space of real numbers, and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function which has a bounded derivative. Prove that f is uniformly continuous on $(\mathbb{R}, |\cdot|)$.

Problem 61. If, in a metric space, a sequence $\{x_n\}_{n \in \mathbb{N}}$ does not converge to y , prove that there exists a open set U containing y and a subsequence of $\{x_n\}_{n \in \mathbb{N}}$ lying outside U .

Problem 62. Let $\{x_n\}_{n \in \mathbb{N}}, y$ be a sequence and a point in a metric space. If every subsequence of $\{x_n\}_{n \in \mathbb{N}}$ has a subsequence converging to y , prove that $\{x_n\}$ converges to y .

Problem 63. Prove that a continuous function from a compact metric space to any metric space is closed (i.e., it sends closed sets into closed sets).

Problem 64. Let f be a continuous one-to-one mapping of a compact metric space (X_1, d_1) onto a metric space (X_2, d_2) . Prove that $f^{-1} : X_2 \rightarrow X_1$ is continuous.

Problem 65. Let A and B be disjoint subsets of a metric space. Suppose that A is closed and B is compact. Prove that the distance between A and B is strictly-positive. Recall distance between A, B is

$$\text{dist}(A, B) = \inf_{x \in A, y \in B} d(x, y)$$

Problem 66. Prove that in a metric space a subset is closed if and only if its intersection with every compact set is closed.

Problem 67. Let (X, d) be a compact metric space. Let A be a set of real functions on M . Assume that A is closed under multiplication and that for any $x \in M$ there exists a function in A vanishing in a neighborhood of x . Prove that A contains the zero function. (The multiplication meant is pointwise: $(fg)(x) = f(x)g(x)$).

Problem 68. Let A and B be subsets of the real numbers. Write $A + B$ for the set of all $a + b$ ($a \in A, b \in B$). If A and B are closed, does it follow that $A + B$ is closed? Does this follow if in addition one of the two is bounded?

Problem 69. Let $(X, |\cdot|)$ be the subspace of the euclidean space, where $X = \{1, \frac{1}{2}, \frac{1}{3}, \dots, 1/n, \dots\}$ (not including 0). Find an open covering of X with no Lebesgue number.

Problem 70. Show that finite union compact sets is a compact set.